

Naive Bayes Thresholds Optuna

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Up to now

- Models (increasingly complex)
- Metrics
- Tuning

Today

- Naive Bayes
- Probability thresholds
- How Optuna works

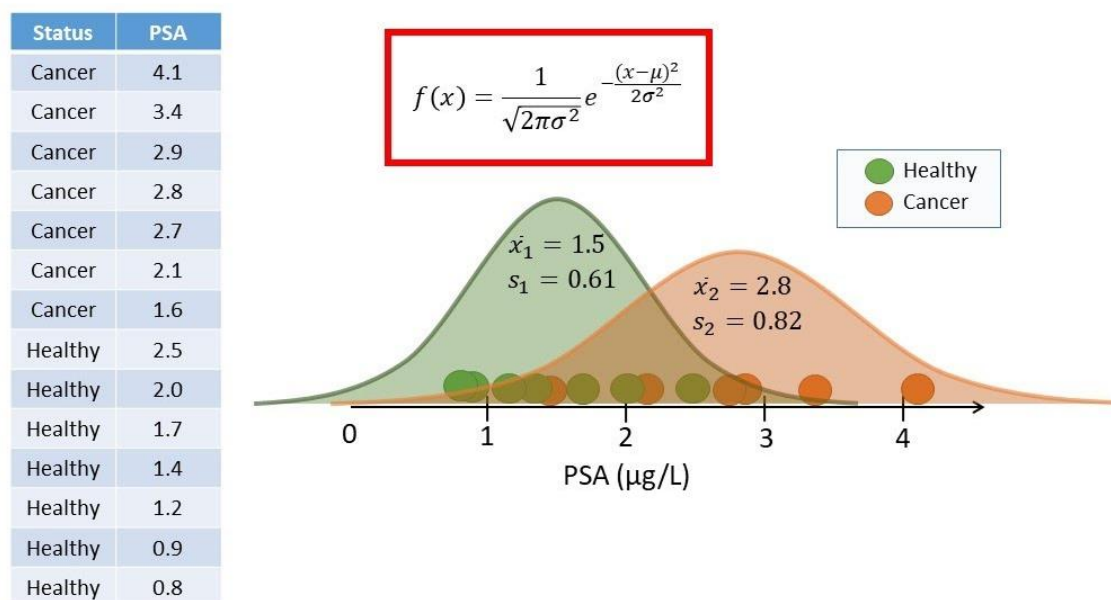
Naive Bayes

$$P(Y | X) = \frac{P(X | Y) \cdot P(Y)}{P(X)}$$

$$P(X_1, X_2, X_3 | Y)$$

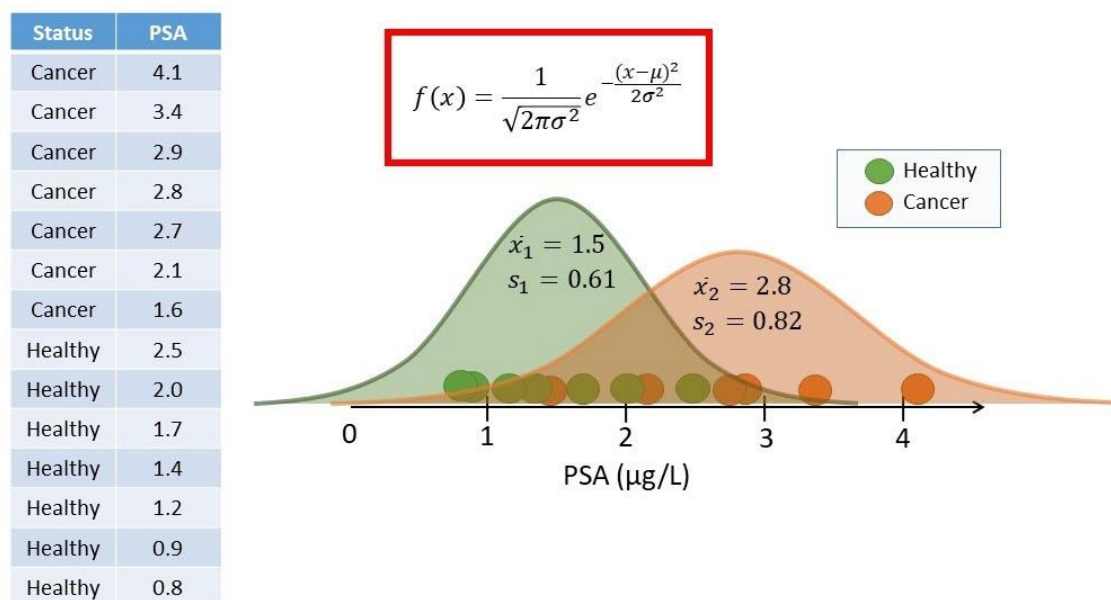
$$P(X | Y) = P(X_1, X_2, X_3, \dots, X_p | Y)$$

GaussianNB



<https://www.youtube.com/watch?v=yRzlyWVEaCQ>

GaussianNB



<https://www.youtube.com/watch?v=yRzlyWVEaCQ>

$$P(Y | X_1, X_2, X_3) \propto P(Y) \cdot P(X_1|Y) \cdot P(X_2|Y) \cdot P(X_3|Y)$$

Other Naive Bayes Models

- GaussianNB
 - For numerical (continuous) features
- BernoulliNB
 - For binary features (0/1)
- Multinomial NB
 - For count data
 - Frequency based features

Thresholds for classification

- If $P(Y=1 | X) \geq 0.5$
 - Then predict class 1
 - Probabilities sum to 1

Changing the threshold

- If $P(Y=1 | X) \geq 0.5$
 - Then predict class 1
 - Probabilities sum to 1

Which metric?

Metric	What it measures	Pros	Cons	When to use
Accuracy	Overall correctness	Simple	Misleading with imbalance	Balanced datasets
Precision	Correctness of positive predictions	Controls false positives	Ignores false negatives	When FP is costly
Recall	Coverage of positives	Finds all positives	Ignores false positives	When FN is costly
F1 Score	Balance of precision & recall	Single number	Hides tradeoff	General summary
Balanced Accuracy	Avg recall per class (0.5=random guess)	Handles imbalance well	Less intuitive	Imbalanced data
ROC AUC	Ranking ability (0.5=random guess)	Threshold-independent	Hard to interpret	Model comparison

Thresholds example

Probability values	Threshold (for class 1)	Predicted Class
$P(Y=1) = 0.7$ $P(Y=0) = 0.3$	0.5	1
$P(Y=1) = 0.7$ $P(Y=0) = 0.3$	0.8	0
$P(Y=1) = 0.7$ $P(Y=0) = 0.3$	0.45	1
$P(Y=1) = 0.4$ $P(Y=0) = 0.6$	0.5	0
$P(Y=1) = 0.4$ $P(Y=0) = 0.6$	0.8	0
$P(Y=1) = 0.4$ $P(Y=0) = 0.6$	0.35	0

How Optuna works

- Tries parameters across your defined ranges
- Uses Bayesian approach to identify which regions work well
- Deeper search in better regions
- Occasionally tries random values to avoid getting stuck